Elena **Marsh**

Research Scientist · Machine Learning for Weather & Energy

CONTACT

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SKILLS

- Probabilistic forecasting
- Python, PyTorch, JAX
- Spatio-temporal deep learning
- Data pipelines (Dask, Zarr, Airflow)
- Cloud and MLOps (AWS, Docker)
- Technical writing and teaching

LANGUAGES

English	native
German	fluent · C1
Spanish	working · B2

LINKS

github.com/emarsh
0000-0002-1825-0097
in/elenamarsh
Google Scholar: E. Marsh

AWARDS

- 2024 RMetS L. F. Richardson Prize
- 2022 Turing Institute Fellowship
- 2020 LMU Dissertation Prize

INTERESTS

Mountain hydrology, open data for public weather services, mentoring first-generation students, long-distance cycling.

PROFILE

Atmospheric scientist turned machine-learning researcher. I build probabilistic forecasting models that turn reanalysis and satellite data into decisions for wind and solar operators, and I care about models that are calibrated, explainable and cheap enough to run every hour. Research posts since 2020 in a university lab and an energy company; looking for a role that keeps one foot in open research.

EXPERIENCE

2023 – now — **Senior Research Scientist**
Northwind Forecasting Ltd, Bristol
Lead a team of four building the company's day-ahead wind power ensemble; cut the 24-hour error by 18% against the previous system and brought inference on 1 200 farms under six minutes.

2020 – 2023 — **Postdoctoral Researcher**
School of Geographical Sciences, University of Bristol
Developed graph neural networks for sub-seasonal precipitation; co-supervised three MSc theses and taught the module *Data Science for the Environment*.

2019 — **Research Intern**
European Centre for Medium-Range Weather Forecasts, Reading
Prototyped neural post-processing of 2 m temperature ensembles; the code fed the centre's first machine-learning pilot.

EDUCATION

2015 – 2020 — **PhD in Atmospheric Physics**
Ludwig-Maximilians-Universität München
Thesis: *Learning the error of convection-permitting forecasts*. Summa cum laude; DWD doctoral fellowship.

2013 – 2015 — **MSc in Physics of the Earth**
University of Edinburgh

SELECTED PROJECTS

skillcurve open-source, 2.1k GitHub stars
A Python library for verifying probabilistic forecasts (CRPS, reliability, sharpness) on labelled arrays; used by two national weather services.

Open Solar Nowcast Turing Institute data challenge, 2022
Organised an open benchmark for 15-minute solar nowcasting from satellite data; 140 teams from 30 countries took part.

SELECTED PUBLICATIONS

- **Marsh, E.**, Okoye, T. and Lund, P. Calibrated graph networks for sub-seasonal rainfall. *Geophysical Research Letters*, 51, 2024.
- **Marsh, E.** and Feld, J. Learning forecast error at the convective scale. *Quarterly Journal of the Royal Meteorological Society*, 147, 2021.
- Haddad, R., **Marsh, E.** *et al.* Ensemble wind power forecasting at fleet scale. *NeurIPS Workshop on Climate Change AI*, 2023.